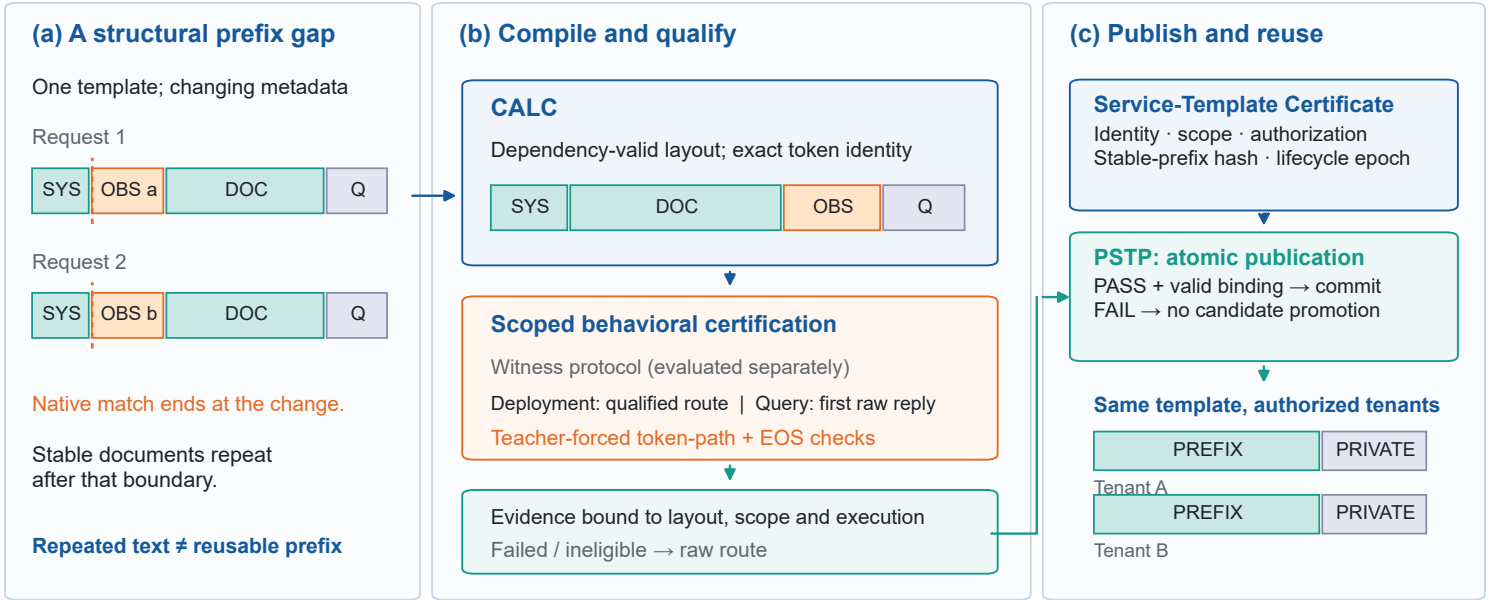
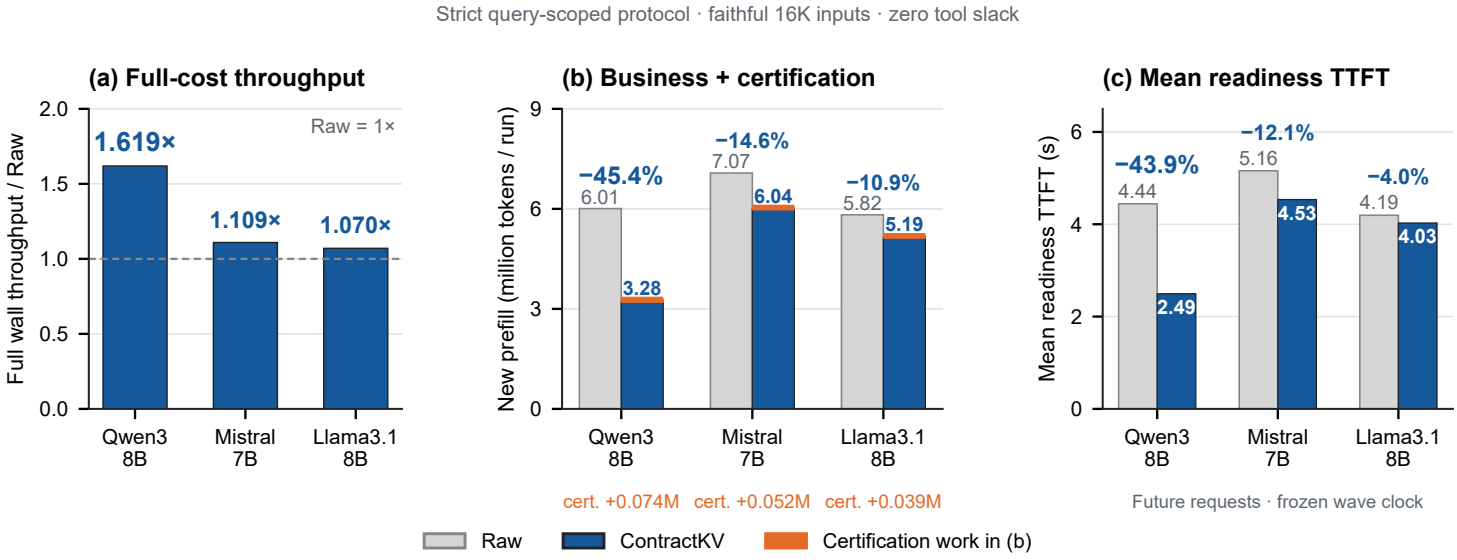
ContractKV | Figure 1

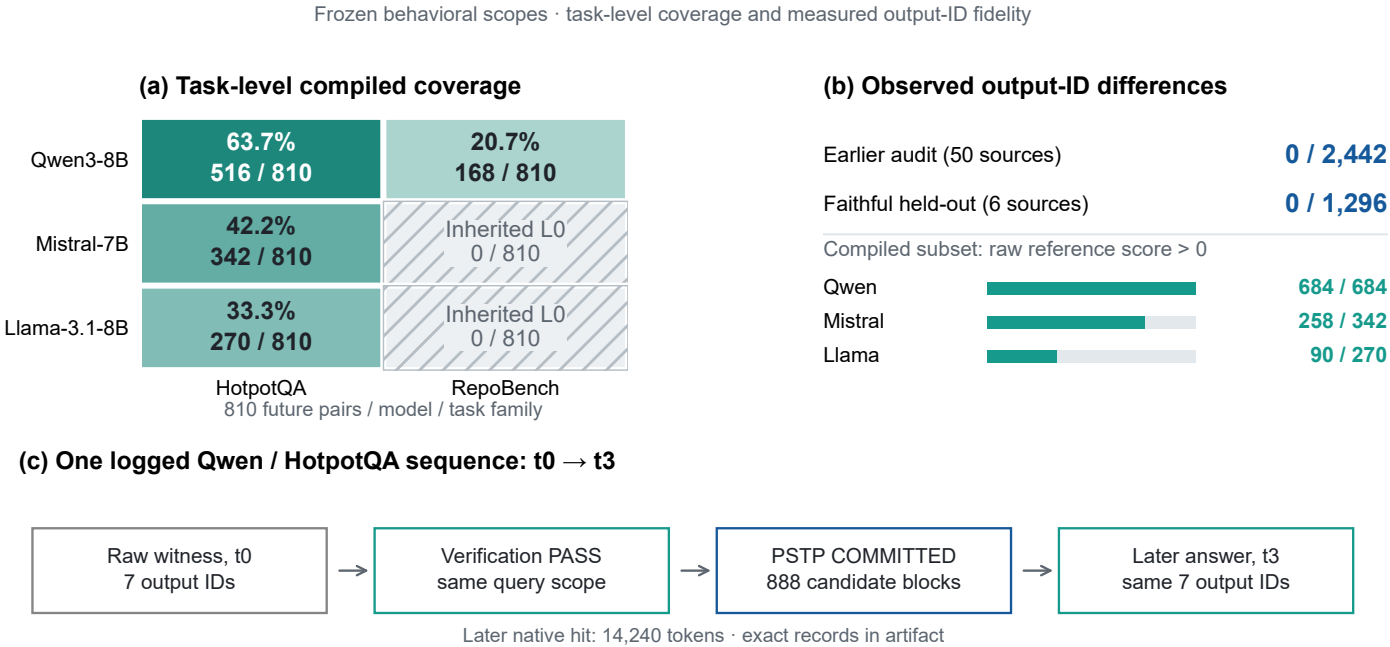


Qualification permits reuse; native cache residency determines the realized hit.

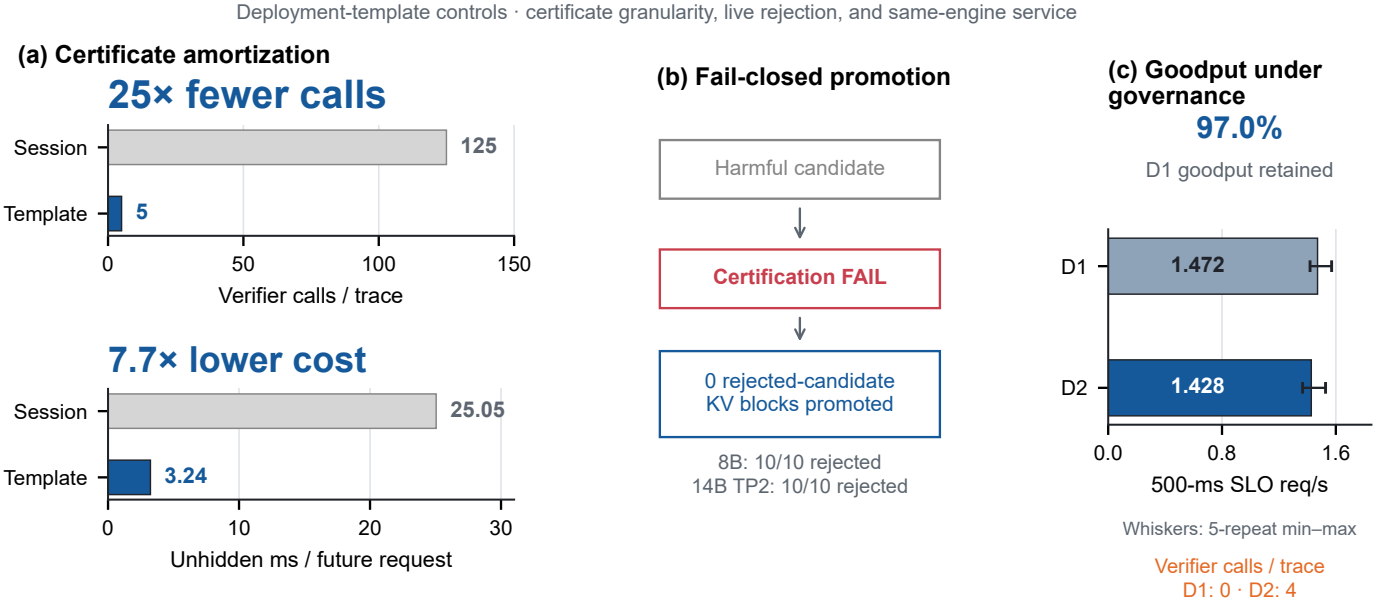
ContractKV makes a compiled stable prefix an evidence-bound service object. CALC repairs a structural prefix gap; one of two separately evaluated witness protocols supplies scoped token-path evidence; PSTP commits eligible state only after PASS and a valid binding. Authorized tenants of the same template share the prefix and keep changing suffixes private. Qualification and physical residency have separate lifetimes.



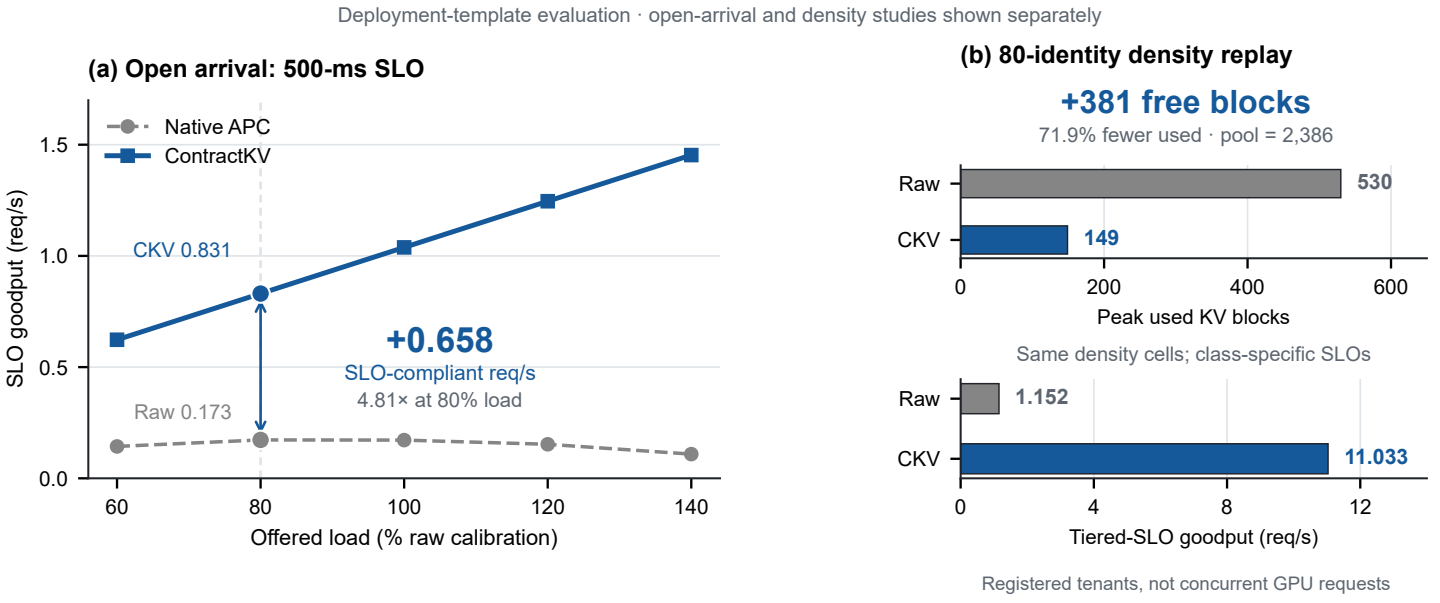
Strict query-qualified performance on six source-disjoint HotpotQA/RepoBench templates, three processes/model, faithful 16K inputs, and zero slack. (a) Throughput charges all business responses, qualification, failed attempts, and synchronous waiting. (b) Actual business plus qualification new-prefill counts per run; orange caps mark certification work at its true scale, with exact added counts below. (c) Mean readiness TTFT over future requests uses the frozen two-request wave clock. Three-repeat values and tail statistics are reported separately.



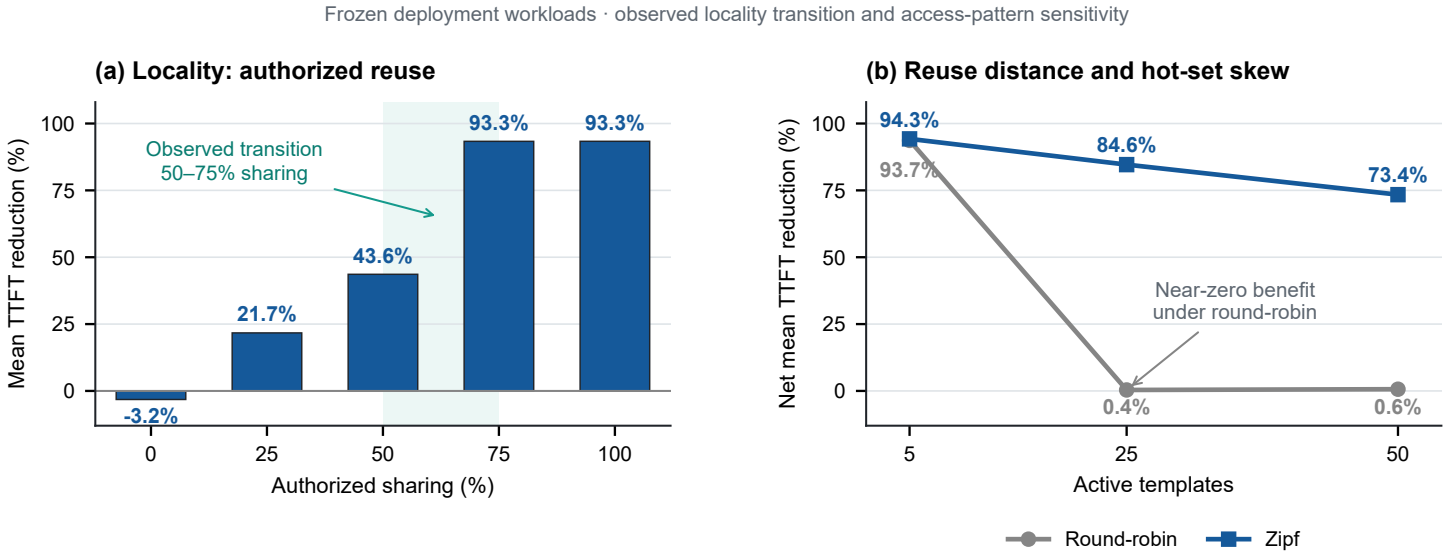
Behavioral evidence with explicit coverage and denominators. (a) Each model/task cell contains 810 future pairs; hatched cells inherit L0 rather than fail GPU certification. (b) The historical five-family audit and faithful primary cohort have zero compiled output-ID differences. Lower bars count raw references receiving positive task-score credit within each model's faithful compiled subset; positive credit is not complete answer correctness. (c) A fixed logged Qwen/HotpotQA sequence links its raw witness, PASS, COMMITTED candidate prefix, and later identical output. Full records are in the artifact.



Deployment-qualified governance and amortization, shown as distinct controls. (a) Holding layout/locality fixed, template registration reduces calls 125 to 5 per 1,125-future-request trace and unhidden cost 25.05 to 3.24 ms/request; three processes. (b) Live 8B and 14B TP2 tests each reject 10/10 harmful dependency-valid candidates, with zero promotion of rejected candidate state. (c) Same-engine D1/D2 mean 500-ms SLO goodput; whiskers denote five-process min-max, not confidence intervals.



Separate deployment-template service studies. (a) Open-arrival 500-ms SLO goodput: at 80% calibrated load, native APC and ContractKV deliver 0.173 and 0.831 requests/s, an absolute gain of 0.658. The 4.81x ratio belongs to that point. (b) In 80-identity density replay, peak used blocks fall 530 to 149 within a 2,386-block pool. The same cells report 1.152/11.033 requests/s under class-specific 750/1,500/3,000-ms SLOs. Registered tenants are not concurrent GPU requests; the two panels use different protocols.



Operating region in frozen deployment workloads. (a) The pale band marks the observed 50-75% sharing interval; the zero-sharing -3.2% result remains visible. (b) Round-robin and Zipf access show how reuse distance and hot-set skew affect net mean TTFT. The axis is measured TTFT reduction, not a direct residency fraction. The band is not a universal threshold.